ACS Energy Letters
http://pubs.acs.org/journal/aelccp

ENERGY EXPRESS

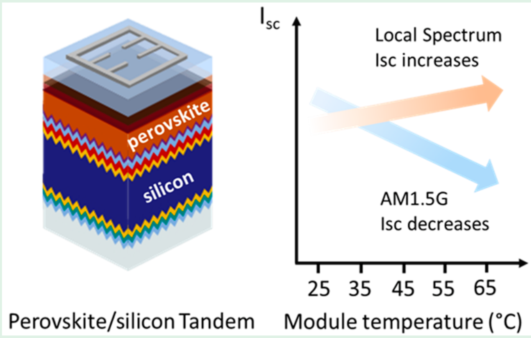
Temperature Coefficients of Perovskite/Silicon Tandem Solar Cells

Maxime Babics,[§] Helen Bristow,[§] Anil Reddy Pininti, Thomas G. Allen, and Stefaan De Wolf*

Cite This: *ACS Energy Lett.* 2023, 8, 3013–3015 Read Online

ACCESS | Metrics & More | Article Recommendations | Supporting Information

ABSTRACT: **Contrasting with single-junction photovoltaic technologies, the short-circuit current temperature coefficient of perovskite/silicon tandem solar cells can be negative, positive, or a mix of both depending on the solar spectrum and operating temperature range.**

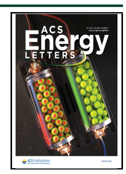


Understanding the outdoor operation of perovskite/silicon tandem solar cells is crucial toward their commercialization. Temperature coefficients (TCs) are key photovoltaic (PV) metrics that quantify the change in current−voltage (*I-V*) parameters per degree centigrade compared to the standard rating at 25 °C.[1] TCs are used to accurately predict energy yields, to monitor *in situ* the performance and stability of PV panels, and to determine the maximum current and voltage generated by a PV installation, thereby dictating the required size of the balance of system components, such as inverters and wires.

TCs of mainstream crystalline silicon (c-Si) PV are well understood and described by a single value for each photovoltaic parameter, valid across typical operation temperatures and illumination intensities.[2] For most PV technologies, the TC of the open-circuit voltage ($TC_{V_{oc}}$) accounts for the main power performance variation. Yet, understanding the short-circuit current temperature coefficient ($TC_{I_{sc}}$) is essential too, for the motives mentioned above. For c-Si, the $TC_{I_{sc}}$ is positive because the silicon bandgap narrows with increasing temperature, resulting in a higher conversion of photons into current.[3] On the contrary, perovskite solar cells have negative $TC_{I_{sc}}$ values since their bandgap widens with increasing temperature.[4,5] The evolution of I_{sc} with temperature for a monolithic perovskite/silicon tandem is less straightforward and depends on the interplay of the subcells' responses to temperature as well as the irradiance spectrum. Earlier, our group first reported on the performance of a perovskite/silicon tandem under elevated temperatures during outdoor operation.[6] This tandem had a negative $TC_{I_{sc}}$ as a result of an increased current mismatch with increased temperature. Yet, in another study, we showed that the current of another tandem installed outdoors displayed a positive $TC_{I_{sc}}$.[7] In this paper, we explain the origin of these seemingly opposing results and demonstrate that, for perovskite/silicon tandems, $TC_{I_{sc}}$ is not a constant value but depends on the initial current mismatch and the local solar spectrum. As a result, $TC_{I_{sc}}$ can be negative, positive, or a mix of both across the operating temperature range.

We fabricated a perovskite/silicon tandem, the device structure of which is shown in Figure 1A. Measured under standard test conditions (STC, 25 °C, AM1.5G spectrum), the tandem has a power conversion efficiency of 29.0% (*I-V* curves provided in Figure S1a). The external quantum efficiency (EQE) of the tandem was measured at device temperatures ranging from 25 to 65 °C, realistic for outdoor operation, as shown in Figure 1B. With increasing temperature, the perovskite bandgap opens and the silicon bandgap narrows, inducing current mismatch (detailed in Figure S1), in agreement with our previous results.[6] To simulate the response of the same tandem at different geographical locations, we generated 5 spectra by changing the blue photon flux of the solar simulator (Figure 1C), thereby generating different levels

Received: May 10, 2023
Accepted: June 13, 2023
Published: June 15, 2023



https://doi.org/10.1021/acsenergylett.3c00930
ACS Energy Lett. 2023, 8, 3013–3015

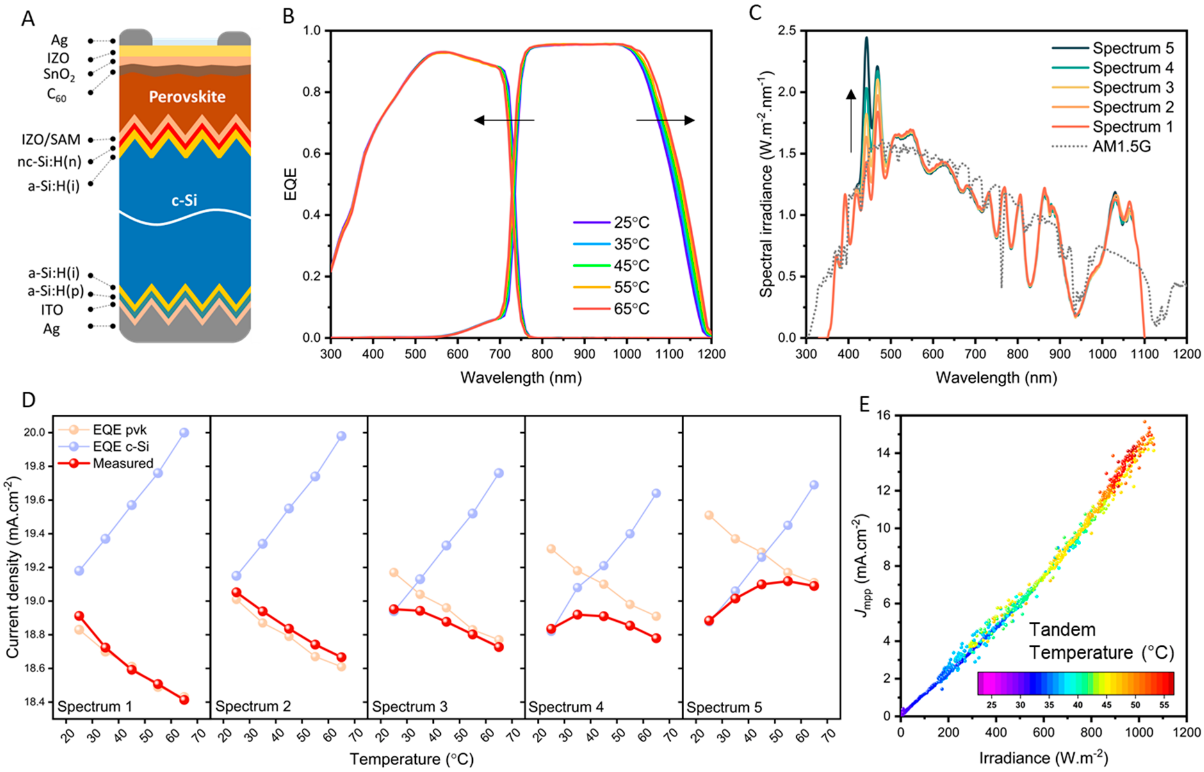


Figure 1. **(A) Perovskite/silicon tandem structure. (B) EQEs of the perovskite/silicon measured at different temperatures. (C) AM1.5G solar spectrum as well as the different spectra used in this study. (D) Theoretical current densities expected from the EQE and measured J_{sc} of the tandem depending on the spectrum at different temperatures. (E) Field-measured J_{mmp} in the outdoor setup of KAUST, Saudi Arabia, depending on the irradiance and cell temperature.**

of current mismatch at 25 °C, where the tandem current is limited by either the perovskite or the silicon subcell (Figure S2).

From the EQEs measured at different temperatures, we calculate the theoretical current density given by each subcell under these spectra (Figure 1D) and distinguish two cases. In the first case, associated with spectra 1 and 2 (characterized by relatively lower blue irradiances), the perovskite subcell is limiting the tandem at all temperatures ≥25 °C. As a result, the tandem has a constant, negative $TC_{I_{sc}}$ under these spectra. From the measured devices and J_{sc}, we calculate $TC_{I_{sc}}$ = −0.051%, in line with the AM1.5G results and our first report.[6]

In the second case, linked with spectra 3, 4, and 5 (characterized by relatively higher blue irradiances), the silicon subcell is limiting the tandem current at 25 °C. This condition can happen in the outdoor environment if the local solar spectrum differs from AM1.5G by having a relatively higher blue photon flux or a lower red photon flux.[7,8] For all the spectra, with increased temperature, the tandem J_{sc} first increases until both subcells generate similar current densities. Beyond this matching temperature T_{match}, the tandem current becomes limited by the perovskite subcell and thus decreases with further temperature increases. In this situation, $TC_{I_{sc}}$ can clearly no longer be described by a single value, and the $TC_{I_{sc}}$ given under STC will not predict the module's behavior during real outdoor operation. The change in I_{sc} with temperature is dictated by the limiting subcell: If the tandem current is perovskite-limited, the $TC_{I_{sc}}$ is negative, but if the tandem current is silicon-limited, the $TC_{I_{sc}}$ is positive below T_{match} and negative above T_{match}. It is worth noting that, for all spectra, the values of $TC_{V_{oc}}$ are similar, around −0.18%, dominating the overall power temperature coefficient $TC_{P_{max}}$ of around −0.21% (Figures S3 and S4). This value is desirably low, lower than that of any single-junction silicon technology.[9]

We confirm this behavior with a tandem installation in our outdoor setup located in KAUST, Saudi Arabia. Under the local spectrum, the tandem is silicon-limited by 2 $mA \cdot cm^{-2}$ (details in Figure S5). Using different days of data collection and plotting the J_{mmp} against irradiance where the data points are indexed on the cell temperature, we confirm that, for a given irradiance, the tandem has a higher current density with increasing temperature, resulting from a reduced current mismatch.[7] These results also imply that careful *in situ* monitoring of $TC_{I_{sc}}$ could give an indication of the amount of current mismatch and could help in understanding degradation and changes in subcell EQEs over time for tandem modules without the need for excessive characterization.

In conclusion, we find that predicting changes in the current of perovskite/silicon tandem solar cells under real outdoor operation is more complex than for single-junction devices. It depends on the EQEs of the tandem subcells, the bandgap of the perovskite, the local solar spectrum, and the operating temperature range. In the case where the tandem cell is silicon-limited, the temperature coefficient is divided into two regimes: positive until T_{match} and negative after T_{match}. These results provide new insights into the perovskite/silicon operation which are essential for accurate energy yield

https://doi.org/10.1021/acsenergylett.3c00930
ACS Energy Lett. 2023, 8, 3013−3015

calculations and the deployment of perovskite/silicon photovoltaic technology.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acsenergylett.3c00930.

> Experimental details, including tandem fabrication, encapsulation, solar cell characterization, and outdoor testing; device performance and EQEs at different temperatures; detailed spectra and the corresponding EQE currents; *I-V* parameters at different temperatures and spectra; *J-V* scans at different temperatures and the temperature coefficients; and the EQE and current densities of an encapsulated tandem (PDF)

AUTHOR INFORMATION

Corresponding Author

Stefaan De Wolf – *KAUST Solar Center, Physical Sciences and Engineering Division (PSE), King Abdullah University of Science and Technology (KAUST), Thuwal 23955-6900, Saudi Arabia;* orcid.org/0000-0003-1619-9061; Email: stefaan.dewolf@kaust.edu.sa

Authors

Maxime Babics – *KAUST Solar Center, Physical Sciences and Engineering Division (PSE), King Abdullah University of Science and Technology (KAUST), Thuwal 23955-6900, Saudi Arabia;* orcid.org/0000-0003-3105-7416

Helen Bristow – *KAUST Solar Center, Physical Sciences and Engineering Division (PSE), King Abdullah University of Science and Technology (KAUST), Thuwal 23955-6900, Saudi Arabia*

Anil Reddy Pininti – *KAUST Solar Center, Physical Sciences and Engineering Division (PSE), King Abdullah University of Science and Technology (KAUST), Thuwal 23955-6900, Saudi Arabia*

Thomas G. Allen – *KAUST Solar Center, Physical Sciences and Engineering Division (PSE), King Abdullah University of Science and Technology (KAUST), Thuwal 23955-6900, Saudi Arabia*

Complete contact information is available at:
https://pubs.acs.org/10.1021/acsenergylett.3c00930

Author Contributions

[§]M.B. and H.B. contributed equally. Conceptualization, M.B. and H.B.; investigation, M.B. and H.B.; resources, A.R.P. and T.G.A.; writing–review and editing, M.B., H.B., T.G.A., and S.D.W.; supervision, S.D.W.

Notes

The authors declare no competing financial interest.

ACKNOWLEDGMENTS

This work was supported by the King Abdullah University of Science and Technology (KAUST) Office of Sponsored Research (OSR) under award nos. OSR-2021-4833, OSR-CARF/CCF-3079, IED OSR-2020-4611, IED OSR-2019-4580, OSR-CRG20204350, OSR-2020-CPF-4519, OSR-CRG2019-4093, and IED OSR-2019-4208. We acknowledge the use of the KAUST Solar Center and the support from its staff.

REFERENCES

(1) Dupré, O.; Vaillon, R.; Green, M. A. Physics of the temperature coefficients of solar cells. *Solar Energy Mater. Solar Cells* **2015**, *140*, 92–100.

(2) Green, M. A. General temperature dependence of solar cell performance and implications for device modelling. *Prog. Photovoltaics: Res. Appl.* **2003**, *11* (5), 333–340.

(3) Green, M. A. Self-consistent optical parameters of intrinsic silicon at 300K including temperature coefficients. *Solar Energy Mater. Solar Cells* **2008**, *92* (11), 1305–1310.

(4) Jošt, M.; Lipovšek, B.; Glažar, B.; Al-Ashouri, A.; Brecl, K.; Matič, G.; Magomedov, A.; Getautis, V.; Topič, M.; Albrecht, S. Perovskite Solar Cells go Outdoors: Field Testing and Temperature Effects on Energy Yield. *Adv. Energy Mater.* **2020**, *10* (25), 2000454.

(5) Moot, T.; Patel, J. B.; McAndrews, G.; Wolf, E. J.; Morales, D.; Gould, I. E.; Rosales, B. A.; Boyd, C. C.; Wheeler, L. M.; Parilla, P. A.; et al. Temperature Coefficients of Perovskite Photovoltaics for Energy Yield Calculations. *ACS Energy Lett.* **2021**, *6* (5), 2038–2047.

(6) Aydin, E.; Allen, T. G.; De Bastiani, M.; Xu, L.; Ávila, J.; Salvador, M.; Van Kerschaver, E.; De Wolf, S. Interplay between temperature and bandgap energies on the outdoor performance of perovskite/silicon tandem solar cells. *Nat. Energy* **2020**, *5* (11), 851–859.

(7) Babics, M.; De Bastiani, M.; Ugur, E.; Xu, L.; Bristow, H.; Toniolo, F.; Raja, W.; Subbiah, A. S.; Liu, J.; Torres Merino, L. V. One-year outdoor operation of monolithic perovskite/silicon tandem solar cells. *Cell Rep. Phys. Sci.* **2023**, *4* (2), 101280.

(8) Kinsey, G. S. Solar cell efficiency divergence due to operating spectrum variation. *Solar Energy* **2021**, *217*, 49–57.

(9) Haschke, J.; Seif, J. P.; Riesen, Y.; Tomasi, A.; Cattin, J.; Tous, L.; Choulat, P.; Aleman, M.; Cornagliotti, E.; Uruena, A.; et al. The impact of silicon solar cell architecture and cell interconnection on energy yield in hot & sunny climates. *Energy Environ. Sci.* **2017**, *10* (5), 1196–1206.